

HOW GUNS WORK

BEFORE THE 19TH CENTURY

Early guns were tubes of bronze or iron, loaded at the muzzle with a propellant (main charge of gunpowder) and a projectile (ball of lead or stone). The barrel had a small hole—a vent, or touchhole—at the breech, into which a user placed priming powder (a small amount of gunpowder). Igniting this priming powder, usually with smoldering match-cord, caused flames to pass down the vent and fire the propellant in the barrel. The vents of later hand-cannon were on the right of the breech, with a shelf, or pan, for the priming powder. Next came devices that ignited the priming powder mechanically. These mechanisms were called locks, because their workings resembled the lock mechanism on a door or chest. The first was the matchlock.



▲ FIRING ARTILLERY

Until the 19th century almost all artillery was fired by match-cord, usually held at the end of a rod (linstock) to allow the gunner to stand away from the recoiling gun. In the late 19th century, gunners were able to fire instantly using “friction tube” primers—copper tubes containing fine gunpowder placed directly into the vent. It was operated by a lanyard, as seen here, which was a length of cord with a hook.

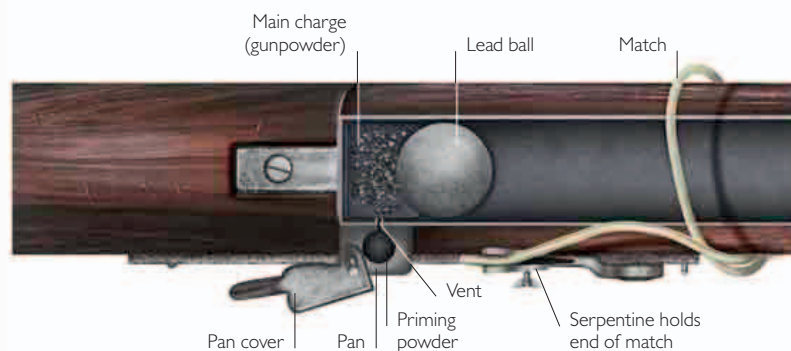


▲ HAND-CANNON

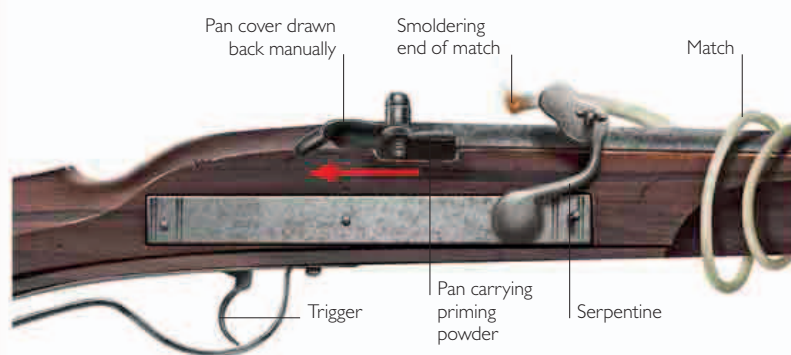
Hand-cannon were the earliest guns small enough to be carried and fired by one user. They had no mechanical firing mechanism—the user touched a smoldering match-cord on the vent manually.

Matchlock

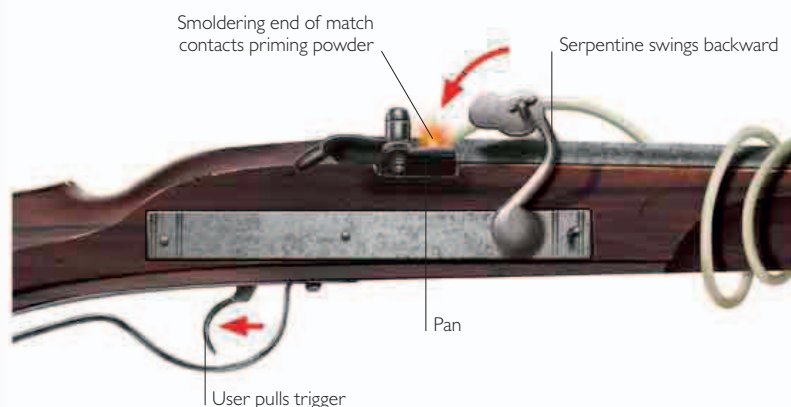
A user loaded a charge of gunpowder and a lead ball at the muzzle, then poured a small amount of finer-grained gunpowder into the priming pan, before closing the pan cover. He would then place a piece of match-cord, its end already smoldering, in the jaws of a snake-shaped match-holder called a serpentine. The user might test the position of the end of the match by gently squeezing the trigger to lower the serpentine, to make sure the match was positioned over the center of the closed pan.



OVERHEAD VIEW OF MATCHLOCK MECHANISM



1 Before firing, the user readies the gun by blowing on the already-smoldering match to enliven it, and by moving the pan cover aside.



2 Pulling the trigger rotates the serpentine, plunging the burning match into the pan with the priming powder. This produces a flash that ignites the main charge via a vent in the side of the barrel.